

BeamContactMapping: Mathematical Formulation

Cosserat Plugin – SOFA Framework

Cosserat Plugin Documentation

Contents

1 Overview

BeamContactMapping is a **Multi2Mapping** that receives the contact-pair descriptors produced by **SphereSweptIntersectionMethod** (SSIM) and converts them into concrete contact-point positions in world space. It also computes the Jacobian relating frame velocities to contact-point velocities, which is required by the constraint solver.

Inputs

- Beam 1 rigid frames: $\{\mathbf{p}_k^{(1)}, \mathbf{R}_k^{(1)}\}_{k=0}^{N_1-1}$ (output of **DiscreteCosseratMapping**).
- Beam 2 rigid frames: $\{\mathbf{p}_k^{(2)}, \mathbf{R}_k^{(2)}\}_{k=0}^{N_2-1}$.
- Per-contact section indices $\{i, j\}$ and curvilinear parameters $\{\alpha, \beta\} \in [0, 1]^2$ from SSIM.
- Beam radii $r_1, r_2 > 0$.

Outputs (for K active contact pairs)

- $2K$ contact-point positions in $\mathbb{R}^3$: index $2k$ is the Beam 1 surface point $\mathbf{P}_A^c$, index $2k + 1$ is the Beam 2 surface point $\mathbf{P}_B^c$.

2 SSIM Output: What the Mapping Receives

For contact pair k , SSIM provides the scalar parameters $(\alpha, \beta) \in [0, 1]^2$ that minimise the surface-to-surface gap.

2.1 ALGO_1 – Segment-to-Segment

$\{i, j\}$ are segment indices. The closest point on Beam 1 is located at parameter α on the segment $[\mathbf{p}_i^{(1)}, \mathbf{p}_{i+1}^{(1)}]$, and the closest point on Beam 2 at parameter β on $[\mathbf{p}_j^{(2)}, \mathbf{p}_{j+1}^{(2)}]$.

2.2 ALGO_2 – Node-to-Segment

i is a node index (not a segment index), j is still a segment index. The Beam 1 contact is located exactly at node $\mathbf{p}_i^{(1)}$, so $\alpha = 0$ by construction. The Beam 2 contact is at parameter β on $[\mathbf{p}_j^{(2)}, \mathbf{p}_{j+1}^{(2)}]$.

3 Forward Map: Contact-Point Positions (apply)

3.1 Centreline Interpolation

ALGO_1

$$\mathbf{P}_A = (1 - \alpha) \mathbf{p}_i^{(1)} + \alpha \mathbf{p}_{i+1}^{(1)}, \quad (1)$$

$$\mathbf{P}_B = (1 - \beta) \mathbf{p}_j^{(2)} + \beta \mathbf{p}_{j+1}^{(2)}. \quad (2)$$

ALGO_2

$$\mathbf{P}_A = \mathbf{p}_i^{(1)} \quad (\text{no interpolation; } \alpha = 0), \quad (3)$$

$$\mathbf{P}_B = (1 - \beta) \mathbf{p}_j^{(2)} + \beta \mathbf{p}_{j+1}^{(2)}. \quad (4)$$

3.2 Contact Normal

$$\hat{\mathbf{n}} = \frac{\mathbf{P}_A - \mathbf{P}_B}{\|\mathbf{P}_A - \mathbf{P}_B\|}. \quad (5)$$

The normal points from Beam 2 centreline toward Beam 1 centreline. If $\|\mathbf{P}_A - \mathbf{P}_B\| < \varepsilon$ (coincident centrelines), a fall-back direction is taken from the local Y -axis of frame i of Beam 1.

3.3 Surface Contact Points

$$\mathbf{P}_A^c = \mathbf{P}_A - r_1 \hat{\mathbf{n}} \quad \rightarrow \text{output}[2k], \quad (6)$$

$$\mathbf{P}_B^c = \mathbf{P}_B + r_2 \hat{\mathbf{n}} \quad \rightarrow \text{output}[2k + 1]. \quad (7)$$

The gap (signed distance between the two surfaces along $\hat{\mathbf{n}}$) is

$$g = \hat{\mathbf{n}} \cdot (\mathbf{P}_B^c - \mathbf{P}_A^c) = \|\mathbf{P}_A - \mathbf{P}_B\| - (r_1 + r_2). \quad (8)$$

Contact is active when $g \leq 0$.

3.4 Radial Moment Arms

Define the moment arm of each contact point relative to its interpolated centreline point:

$$\mathbf{a}_A = \mathbf{P}_A^c - \mathbf{P}_A = -r_1 \hat{\mathbf{n}}, \quad \mathbf{a}_B = \mathbf{P}_B^c - \mathbf{P}_B = +r_2 \hat{\mathbf{n}}. \quad (9)$$

A key property used in the Jacobian derivation (Section ??) is that $\mathbf{a}_A$ and $\mathbf{a}_B$ are the *same* vector for all frames contributing to the same contact point, because the SLERP linearisation collapses per-frame moment variations into a single radial term.

4 Tangent Map: Velocity Jacobian (applyJ)

4.1 Rigid-Frame Velocity Representation

Each Cosserat beam frame carries a 6-DOF velocity: $\mathbf{d}_k = (\mathbf{v}_k, \boldsymbol{\omega}_k)$ where $\mathbf{v}_k \in \mathbb{R}^3$ is the linear velocity and $\boldsymbol{\omega}_k \in \mathbb{R}^3$ is the angular velocity of frame k .

The velocity of any point $\mathbf{q}$ rigidly attached to frame k is

$$\dot{\mathbf{q}} = \mathbf{v}_k + \boldsymbol{\omega}_k \times (\mathbf{q} - \mathbf{p}_k). \quad (10)$$

4.2 Velocity of the Centreline Contact Point

Differentiating the linear interpolation (??) and applying the SLERP linearisation (angular velocity interpolates linearly to first order in α):

$$\dot{\mathbf{P}}_A = (1 - \alpha) \mathbf{v}_i + \alpha \mathbf{v}_{i+1} \quad (\text{ALGO_1}). \quad (11)$$

For ALGO_2 ($\alpha = 0$):

$$\dot{\mathbf{P}}_A = \mathbf{v}_i. \quad (12)$$

4.3 Velocity of the Surface Contact Point

Differentiating (??) with the contact normal $\hat{\mathbf{n}}$ frozen at the current configuration:

$$\dot{\mathbf{P}}_A^c = \dot{\mathbf{P}}_A - r_1 \dot{\hat{\mathbf{n}}} \approx \dot{\mathbf{P}}_A + r_1 [(1 - \alpha) \boldsymbol{\omega}_i + \alpha \boldsymbol{\omega}_{i+1}] \times \hat{\mathbf{n}}. \quad (13)$$

Factoring by frame:

$$\dot{\mathbf{P}}_A^c = \sum_{b \in \mathcal{B}_1(k)} w_b [\mathbf{v}_b + \boldsymbol{\omega}_b \times \mathbf{a}_A], \quad (14)$$

where $\mathcal{B}_1(k)$ and the weights w_b are defined in Table ??, and $\mathbf{a}_A = -r_1 \hat{\mathbf{n}}$.

Similarly for Beam 2:

$$\dot{\mathbf{P}}_B^c = \sum_{b \in \mathcal{B}_2(k)} w_b [\mathbf{v}_b + \boldsymbol{\omega}_b \times \mathbf{a}_B], \quad (15)$$

with $\mathcal{B}_2(k)$ and $\mathbf{a}_B = +r_2 \hat{\mathbf{n}}$.

Table 1: Jacobian block sets for contact pair k .

Algorithm	Surface pt	Block set	Frame index	Weight
ALGO_1	$\mathbf{P}_A^c$	$\mathcal{B}_1(k)$	i	$(1 - \alpha)$
			$i + 1$	α
ALGO_2	$\mathbf{P}_A^c$	$\mathcal{B}_1(k)$	i only	1
Both	$\mathbf{P}_B^c$	$\mathcal{B}_2(k)$	j	$(1 - \beta)$
			$j + 1$	β

4.4 Compact Jacobian Notation

Stack the frame velocities of Beam 1 into $\dot{\mathbf{q}}_1 \in \mathbb{R}^{6N_1}$ and Beam 2 into $\dot{\mathbf{q}}_2 \in \mathbb{R}^{6N_2}$. The contact-point velocity $\dot{\mathbf{s}}_k \in \mathbb{R}^3$ at output DOF $2k$ satisfies

$$\dot{\mathbf{s}}_k = J_k^{(1)} \dot{\mathbf{q}}_1, \quad J_k^{(1)} \in \mathbb{R}^{3 \times 6N_1}, \quad (16)$$

where the only non-zero 3×6 blocks of $J_k^{(1)}$ are

$$\left[J_k^{(1)} \right]_{\cdot, b} = w_b \left[I_{3 \times 3} \mid -[\mathbf{a}_A]_{\times} \right], \quad b \in \mathcal{B}_1(k), \quad (17)$$

with $[\mathbf{a}_A]_\times$ the skew-symmetric matrix associated with $\mathbf{a}_A$ (so that $[\mathbf{a}_A]_\times \boldsymbol{\omega} = \mathbf{a}_A \times \boldsymbol{\omega}$).

Analogously, for output DOF $2k + 1$:

$$\left[J_k^{(2)} \right]_{\cdot, b} = w_b \left[I_{3 \times 3} \mid -[\mathbf{a}_B]_\times \right], \quad b \in \mathcal{B}_2(k). \quad (18)$$

5 Transpose Map: Force Back-Propagation (applyJT)

5.1 Virtual Work Principle

By the principle of virtual work, a contact force $\mathbf{F}$ applied at $\mathbf{P}_A^c$ is dual to the velocity $\dot{\mathbf{P}}_A^c$:

$$\delta W = \mathbf{F} \cdot \delta \mathbf{P}_A^c = \sum_{b \in \mathcal{B}_1(k)} w_b \mathbf{F} \cdot [\delta \mathbf{p}_b + \delta \boldsymbol{\theta}_b \times \mathbf{a}_A]. \quad (19)$$

Identifying the conjugate generalised forces:

$$\boxed{\mathbf{f}_b += w_b \mathbf{F}, \quad \boldsymbol{\tau}_b += w_b (\mathbf{a}_A \times \mathbf{F}), \quad b \in \mathcal{B}_1(k).} \quad (20)$$

For Beam 2, a force $\mathbf{F}'$ at $\mathbf{P}_B^c$ gives:

$$\boxed{\mathbf{f}_b += w_b \mathbf{F}', \quad \boldsymbol{\tau}_b += w_b (\mathbf{a}_B \times \mathbf{F}'), \quad b \in \mathcal{B}_2(k).} \quad (21)$$

5.2 Normal vs. Tangential Forces

Normal contact force. $\mathbf{F} = \lambda \hat{\mathbf{n}}$ for some $\lambda \geq 0$:

$$\boldsymbol{\tau}_b = w_b (\mathbf{a}_A \times \lambda \hat{\mathbf{n}}) = w_b (-r_1 \hat{\mathbf{n}}) \times (\lambda \hat{\mathbf{n}}) = -w_b r_1 \lambda (\hat{\mathbf{n}} \times \hat{\mathbf{n}}) = \mathbf{0}. \quad (22)$$

A pure normal force produces *no torque* on any frame. Physically: the force is directed along the line connecting the frame centre to the contact point, so its moment arm is zero.

Tangential (friction) force. Let $\hat{\mathbf{t}} \perp \hat{\mathbf{n}}$ be a tangential direction. $\mathbf{F} = \lambda_t \hat{\mathbf{t}}$:

$$\boldsymbol{\tau}_b = w_b (-r_1 \hat{\mathbf{n}}) \times (\lambda_t \hat{\mathbf{t}}) = -w_b r_1 \lambda_t (\hat{\mathbf{n}} \times \hat{\mathbf{t}}) \neq \mathbf{0}. \quad (23)$$

A tangential force generates a rolling/twisting torque on the frames.

6 Constraint Jacobian (applyJT MatrixDeriv)

6.1 Role in the Constraint Solver

GenericConstraintSolver (under **FreeMotionAnimationLoop**) assembles a constraint Jacobian H by calling **applyJT** on each mapping in the scene graph. Each *constraint* corresponds to one scalar condition $g(\mathbf{q}) = 0$, evaluated along a direction $\mathbf{d} \in \mathbb{R}^3$ at a specific contact point.

A row r of the output constraint matrix encodes:

$$[\text{OutMatrixDeriv}]_{r, \text{outDOF}} = \mathbf{d} \in \mathbb{R}^3. \quad (24)$$

For the normal contact constraint, $\mathbf{d} = \hat{\mathbf{n}}$ and $\text{outDOF} \in \{2k, 2k + 1\}$ corresponds to one of the two surface points of pair k .

6.2 Mapping to Input Space

The corresponding contribution to the input (frame) constraint matrix is:

For $\text{outDOF} = 2k$ (Beam-1 surface point, blocks $b \in \mathcal{B}_1(k)$):

$$[\text{In1MatrixDeriv}]_{r,b,\text{frameIdx}} += \begin{pmatrix} w_b \mathbf{d} \\ w_b (\mathbf{a}_A \times \mathbf{d}) \end{pmatrix} \in \mathbb{R}^6. \quad (25)$$

For $\text{outDOF} = 2k + 1$ (Beam-2 surface point, blocks $b \in \mathcal{B}_2(k)$):

$$[\text{In2MatrixDeriv}]_{r,b,\text{frameIdx}} += \begin{pmatrix} w_b \mathbf{d} \\ w_b (\mathbf{a}_B \times \mathbf{d}) \end{pmatrix} \in \mathbb{R}^6. \quad (26)$$

6.3 Normal Constraint Row (Explicit Form)

The gap rate for contact k along $\hat{\mathbf{n}}$ is

$$\dot{g} = \hat{\mathbf{n}} \cdot (\dot{\mathbf{P}}_B^c - \dot{\mathbf{P}}_A^c). \quad (27)$$

Substituting (??)–(??):

$$\dot{g} = \hat{\mathbf{n}} \cdot \left[\sum_{b \in \mathcal{B}_2(k)} w_b (\mathbf{v}_b + \boldsymbol{\omega}_b \times \mathbf{a}_B) - \sum_{b \in \mathcal{B}_1(k)} w_b (\mathbf{v}_b + \boldsymbol{\omega}_b \times \mathbf{a}_A) \right]. \quad (28)$$

Using $\hat{\mathbf{n}} \cdot (\boldsymbol{\omega} \times (-r_1 \hat{\mathbf{n}})) = -r_1 \hat{\mathbf{n}} \cdot (\boldsymbol{\omega} \times \hat{\mathbf{n}}) = 0$ (the scalar triple product $[\hat{\mathbf{n}}, \hat{\mathbf{n}}, \boldsymbol{\omega}] = 0$), the angular-velocity contributions to the *normal* constraint vanish:

$$\dot{g} = \sum_{b \in \mathcal{B}_2(k)} w_b \hat{\mathbf{n}} \cdot \mathbf{v}_b - \sum_{b \in \mathcal{B}_1(k)} w_b \hat{\mathbf{n}} \cdot \mathbf{v}_b, \quad (29)$$

confirming that the 6-DOF Jacobian blocks for the normal constraint reduce to:

$$[J_k^{(1)}]_{\text{normal},b} = [(1 - \alpha) \hat{\mathbf{n}}^T \mid \mathbf{0}_{1 \times 3}], \quad [J_k^{(2)}]_{\text{normal},b} = [(1 - \beta) \hat{\mathbf{n}}^T \mid \mathbf{0}_{1 \times 3}]. \quad (30)$$

6.4 Tangential Constraint Row (Explicit Form)

For a tangential direction $\hat{\mathbf{t}} \perp \hat{\mathbf{n}}$, the angular-velocity terms do not cancel. Using the cyclic identity $\hat{\mathbf{t}} \cdot (\boldsymbol{\omega} \times \mathbf{a}) = \boldsymbol{\omega} \cdot (\mathbf{a} \times \hat{\mathbf{t}})$:

$$[J_k^{(1)}]_{\text{tan},b} = w_b [\hat{\mathbf{t}}^T \mid (\mathbf{a}_A \times \hat{\mathbf{t}})^T] = w_b [\hat{\mathbf{t}}^T \mid (-r_1 \hat{\mathbf{n}} \times \hat{\mathbf{t}})^T]. \quad (31)$$

The rotational block $-r_1 (\hat{\mathbf{n}} \times \hat{\mathbf{t}})$ is the moment-arm coupling that links frame rotation to tangential slip.

7 ALGO_1 vs. ALGO_2: Summary of Differences

ALGO_2 can be seen as the degenerate limit of ALGO_1 with $\alpha = 0$: the weight of frame $i + 1$ vanishes, and the mapping needs only one Beam-1 Jacobian block. The same formulae (??)–(??) apply in both cases without special-casing – only the block set $\mathcal{B}_1(k)$ and the interpretation of the index i differ.

Table 2: Key differences between ALGO_1 and ALGO_2.

Property	ALGO_1 (segment-to-segment)	ALGO_2 (node-to-segment)
Index i	Beam-1 <i>segment</i> index	Beam-1 <i>node</i> index
Parameter α	$\in (0, 1)$ in general	$= 0$ always
Beam-1 blocks per contact	2 (frames i and $i + 1$)	1 (frame i only)
Weights on Beam-1	$(1 - \alpha), \alpha$	1
Beam-2 blocks per contact	2 (frames j and $j + 1$), weights $(1 - \beta)$ and β	
Output positions per contact	2 (Pc_A at $2k$, Pc_B at $2k + 1$)	

8 Symbol Glossary

Symbol	Meaning
$\mathbf{p}_k^{(m)}$	Position of frame k on Beam m
$\mathbf{R}_k^{(m)}$	Orientation (rotation matrix) of frame k on Beam m
N_m	Number of frames on Beam m
i	Beam-1 segment index (ALGO_1) or node index (ALGO_2)
j	Beam-2 segment index
α, β	Curvilinear parameters $\in [0, 1]$ from SSIM
r_1, r_2	Cross-section radii of Beam 1 and Beam 2
$\mathbf{P}_A, \mathbf{P}_B$	Interpolated centreline contact points
$\hat{\mathbf{n}}$	Unit contact normal (Beam-2 $\rightarrow$ Beam-1)
$\mathbf{P}_A^c, \mathbf{P}_B^c$	Surface contact points
g	Signed gap ($= 0$ at first contact, < 0 in penetration)
$\mathbf{a}_A, \mathbf{a}_B$	Radial moment arms: $-r_1\hat{\mathbf{n}}$ and $+r_2\hat{\mathbf{n}}$
$\mathbf{v}_k, \boldsymbol{\omega}_k$	Linear and angular velocities of frame k
w_b	Interpolation weight of Jacobian block b
$\mathbf{F}$	Contact force vector
$\mathbf{f}_b, \boldsymbol{\tau}_b$	Generalised force and torque accumulated on frame b
K	Number of active contact pairs
ε	Numerical zero threshold (10^{-14})